

SinhalaJournal-LLM: A Style-Controlled Large Language Model for Diverse Sri Lankan Newspaper Writing

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Project Proposal Report

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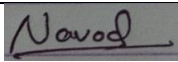
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

Brief background of the problem: Sinhala is a low-resource language with limited Natural Language Processing (NLP) tools, especially for news text summarization. Most available summarizers for Sinhala use extractive methods that select sentences based on keyword frequency or ranking algorithms but often fail to produce fluent or semantically coherent summaries of long news articles [6].

Identified research gap: Existing research for Sinhala summarization mainly focuses on extractive techniques and shallow statistical methods. There is no robust abstractive news summarization system that generates human-like summaries while maintaining specific journalistic tones and preventing AI hallucinations. Moreover, recent advances in Large Language Models (LLMs) like SinLlama [6] show improved language understanding but have not yet been applied specifically for style-aware summarization tasks.

Proposed solution overview: This research proposes an AI-powered Sinhala News Summarizer that generates concise, meaning-preserving abstractive summaries. Acting as a core component of the SinhalaJournal-LLM toolkit, the system utilizes dynamic Low-Rank Adaptation (LoRA) to match specific journalistic styles (e.g., sports, editorial). It also integrates an internal factual consistency filter to ensure journalistic integrity before routing the output for grammar validation.

Technologies to be used: The summarizer will use a Sinhala-adapted LLM architecture, specifically the foundational SinLlama model [6], augmented with Parameter-Efficient Fine-Tuning (PEFT) techniques [6]. Custom tokenization and context window management will be applied to process lengthy Sinhala news corpora efficiently within hardware constraints.

Expected outcomes: The summarization component is expected to significantly outperform extractive baselines on automatic evaluation metrics such as ROUGE [6] and semantic similarity scores [6], alongside modern "LLM-as-a-judge" qualitative evaluations. The system will deliver fluent, stylistically accurate, and factually reliable summaries.

Social and commercial impact: Socially, the system will improve accessibility to Sinhala news by reducing reading time for the public. Commercially, it provides a viable B2B solution for Sri Lankan media platforms to eliminate editorial bottlenecks, drastically reduce manual labor costs, and enhance digital user engagement.

Keywords: Sinhala NLP, Abstractive Summarization, Large Language Models, Parameter-Efficient Fine-Tuning, Journalism AI.

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Special thanks to the developers and contributors of the SinLlama and related Sinhala NLP research projects for their open research efforts, which laid the foundation for advancing NLP tools in low-resource languages.

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LIST OF ABBREVIATIONS

- **API** – Application Programming Interface
- **AWS** – Amazon Web Services
- **B2B** – Business-to-Business
- **B2C** – Business-to-Consumer
- **BPE** – Byte Pair Encoding
- **CMS** – Content Management System
- **DSR** – Design Science Research
- **GPU** – Graphics Processing Unit
- **IFT** – Instruction Fine-Tuning
- **IP** – Intellectual Property
- **LCS** – Longest Common Subsequence
- **LLM** – Large Language Model
- **LoRA** – Low-Rank Adaptation
- **NLP** – Natural Language Processing
- **PEFT** – Parameter-Efficient Fine-Tuning
- **ROI** – Return on Investment
- **ROUGE** – Recall-Oriented Understudy for Gisting Evaluation
- **SaaS** – Software-as-a-Service
- **SDG** – Sustainable Development Goals
- **UI** – User Interface
- **VRAM** – Video Random Access Memory
- **WBS** – Work Breakdown Structure

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1. INTRODUCTION

1.1. Background and Context

Many Sinhala readers struggle to quickly grasp the meaning of long news articles due to lack of summarization tools tailored for Sinhala news [2].

Readers, journalists, students, and news platforms are affected. Without reliable summaries, information access is slower and less efficient [1] [2].

Global news consumption is increasing rapidly, with over 4.6 billion internet users accessing news online yearly. However, Sinhala summarization tools are extremely limited compared with English systems [2].

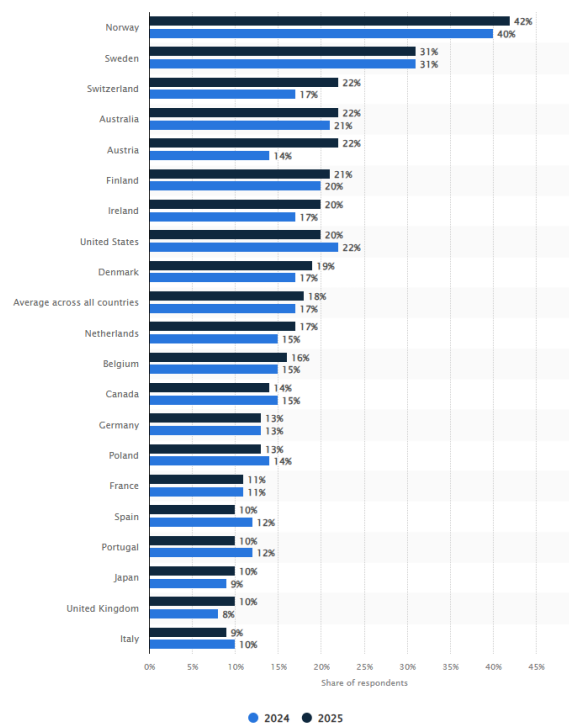


Figure 1-Share of consumers who paid for online news content in selected countries worldwide in the last year in 2024 and 2025

Existing solutions use extractive methods (e.g. TF-IDF, TextRank) that select sentences but do not capture meaning or produce fluent summaries [2] [5]. These also lack deep semantic understanding.

We propose an *abstractive news summarizer* using a Sinhala-trained LLM that:

- understands context and meaning,
- generates concise, fluent summaries,
- adapts to journalistic language.

Supports SDG 4 (Quality Education) by improving access to information and reading comprehension for Sinhala speakers.

Aligned with Natural Language Processing, Machine Learning for Low-Resource Languages, and Human–Computer Interaction.

1.2.Literature Review

Recent research on automatic text summarization spans both general NLP developments and specific efforts for low-resource languages like Sinhala.

SinDOC introduces a hybrid summarization technique combining extractive and abstractive methods for Sinhala documents and reports that the combined approach outperforms purely extractive systems in preliminary evaluations [6]. This work highlights the necessity of incorporating semantic understanding into summaries rather than relying solely on sentence selection.

The SinLlama project demonstrates how Large Language Models (LLMs) can be specially adapted for Sinhala by improving tokenization and continual pre-training on a large corpus of Sinhala text [1]. Although not directly applied to summarization, this research provides an essential foundation for semantic tasks such as news summarization, showing that LLMs fine-tuned on Sinhala outperform generic multilingual models.

Earlier work such as Automatic Text Summarization for Sinhala examines extractive summarization techniques using linguistic and statistical approaches like TextRank and TF-IDF to select salient sentences from text [7]. While valuable as a benchmark, such extractive methods cannot paraphrase or combine ideas into fluent summaries, limiting their usefulness for more natural, human-like summaries.

Broader research into abstractive summarization methodologies highlights the importance of encoder–decoder architectures and sequence-to-sequence learning for generating summaries that are semantically richer than extractive outputs [8]. This reflects trends where transformer-based models greatly improve summary coherence and quality.

Studies on abstractive summarization for other low-resource languages (e.g., Urdu) illustrate that deep learning approaches can surpass traditional statistical baselines when trained on even modest amounts of language-specific data. These findings suggest that similar strategies could be beneficial for Sinhala if enough training resources are available.

Focused analyses of Sinhala online article summarization show that current extractive baselines like TF-IDF and TextRank yield moderate performance but struggle with semantic meaning representation, further motivating research into abstractive systems [2]. Additionally, Talking Books explores bilingual abstractive and extractive algorithms specifically for Sinhala text, reinforcing the research community’s interest in moving beyond sentence extraction.

Finally, large multilingual resources like XL-Sum provide multilingual abstractive summarization datasets covering dozens of languages, setting useful baselines and transferable architectures that could inform a Sinhala summarizer’s design [9].

Summary of Approaches:

- Extractive models (TF-IDF, TextRank) identify key sentences but cannot generate new phrasing [7] [6].

- Hybrid methods combine extractive ranking with abstractive components to improve summary content [9].
- Abstractive models using deep learning or LLMs generate semantically coherent text and are preferred for natural summaries [8] [9] [10].
- Language-specific adaptations (SinLlama) show promise for low-resource language tasks but need task-specific fine-tuning [1].

Limitations of Existing Systems:

Despite progress, current Sinhala summarization research remains limited in data volume, linguistic resources, and integration with modern LLM techniques. Extractive systems lack fluency and semantic richness, while abstractive approaches for Sinhala are at an early stage without robust evaluations on diverse news corpora. Furthermore, multilingual summarization resources provide models but not language-specific fine-tuning for Sinhala news text.

1.3. Research Gap

Existing studies in Sinhala text summarization primarily focus on extractive methodologies. Researchers have extensively utilized classical, statistical, and graph-based algorithms—such as TF-IDF, TextRank, and feature weight algorithms—to extract the most salient sentences directly from source documents [7] [2]. While these extractive techniques are computationally lightweight and ensure the exact words of the original text are retained, they fundamentally lack the ability to paraphrase contextually or adapt to different writing styles [7].

In the realm of abstractive summarization, recent research has explored fine-tuning multilingual sequence-to-sequence models (such as mT5 or mBART) utilizing datasets like XL-Sum [9] [12]. However, empirical studies highlight that general multilingual Large Language Models (multiLLMs) often underperform on low-resource languages with complex morphology, frequently suffering from model hallucination and factual inconsistency [8]. Most importantly, these existing abstractive systems treat all text uniformly and lack structural style control. Current systems cannot detect or replicate the distinct writing styles required in journalism—such as the formal, analytical tone necessary in editorials versus the condensed, pressing style of breaking news.

The Knowledge Gap: Existing studies focus on general-purpose extractive summarization or basic multilingual abstractive generation for the Sinhala language. However, none address the generation of abstractive summaries that actively preserve and reflect the specific *journalistic style* of the original Sinhala news articles (e.g., sports, editorial, political) using advanced Large Language Models.

Novelty and Creativity: The proposed Sinhala News Summarizer addresses this gap by introducing a style-aware abstractive summarization approach integrated into the SinhalaJournal-LLM framework. Moving beyond rigid sentence-extraction techniques, this solution leverages a foundational LLM architecture (such as SinLlama) fine-tuned on locally sourced Sri Lankan news data [1]. This approach is highly novel as it not only condenses lengthy Sinhala news articles but specifically decodes the contextual tone of the input data to generate high-quality summaries that dynamically mimic the original article's distinct journalistic style.

2. OBJECTIVE

2.1. Main Objective

Specific: Clearly defines what is being built (a style-aware abstractive summarizer), who it is for (Sinhala journalism), and how it will be built (using SinLlama and domain-specific LoRA adapters).

Measurable: Defines exactly how success will be tested using standard NLP evaluation metrics (ROUGE scores) and implies benchmarking against current standards [3] [8].

Achievable: Uses established parameter-efficient fine-tuning techniques (LoRA) on an existing foundational model (SinLlama/Llama-3), ensuring the project can be completed with limited computational resources [1].

Relevant: Directly solves the knowledge gap established in Section 1.3 by providing a tool that does not just extract sentences, but understands and replicates journalistic tone (e.g., political, sports, editorial) for the Sinhala language.

Time-Bound: Acknowledges the academic constraint of the research taking place over the standard undergraduate project timeline (two semesters)

2.2. Specific Objectives

1. To curate and pre-process a domain-specific Sinhala news dataset: To collect, clean, and format a high-quality dataset of Sinhala news articles and their corresponding reference summaries, categorized by specific journalistic domains (e.g., Political, Sports, Editorial), and to configure a custom Byte-Level BPE tokenizer to accurately map Sinhala vocabulary to the base LLM.
2. To implement style-aware Parameter-Efficient Fine-Tuning (PEFT): To train domain-specific Low-Rank Adaptation (LoRA) adapters on top of the foundational SinLlama model, enabling the architecture to capture and dynamically replicate the distinct writing styles, tones, and vocabulary of different news categories without catastrophic forgetting.
3. To develop the style-aware abstractive generation module: To design an optimized prompt-wrapper mechanism that guides the fine-tuned LLM to accurately condense lengthy Sinhala news articles into concise abstractive summaries that strictly reflect the selected journalistic domain's style.
4. To integrate a post-processing refinement loop: To engineer a feedback and refinement module that validates the generated Sinhala summaries for factual consistency and grammatical correctness, actively mitigating model hallucination and ensuring high-quality, reliable journalistic outputs.
5. To evaluate summarization performance using standard NLP metrics: To quantitatively and qualitatively validate the system's accuracy and style retention by comparing the machine-generated summaries against human-generated reference summaries, primarily utilizing ROUGE metrics (ROUGE-1, ROUGE-2, and ROUGE-L) and LLM-as-a-Judge semantic evaluations.

3. METHODOLOGY

3.1. Research Approach

This study adopts the Design Science Research (DSR) methodology. DSR is highly suitable for this project because the core contribution is the creation of a novel IT artifact—specifically, an AI-powered, style-aware abstractive text summarization model tailored for the low-resource Sinhala language. Rather than merely analyzing data or comparing existing algorithms, this methodology involves the structured design, development, and rigorous evaluation of a working system intended to improve operational efficiency in a specific domain (Sri Lankan newsrooms). The DSR framework will be executed in iterative cycles: understanding the editorial problem, designing the LLM architecture, developing the artifact, and conducting structured evaluations to prove its real-world applicability.

3.2. Data Collection and Ethical Considerations

- **Data Collection:** The primary dataset will be entirely custom-curated by the research team using multiple automated web scrapers built for different reputable Sri Lankan news websites, including Lankadeepa, Hiru News, Ada, and Divaina. This proprietary scraping pipeline will extract structured news articles containing the full article bodies, headlines, publication dates, and category information. The collected data will then be carefully categorized into distinct journalistic domains (e.g., Political, Sports, Editorial) to support the model's style-aware training.
- **Pre-processing:** Raw scraped text will undergo rigorous cleaning. This includes duplicate removal, length-based filtering to exclude text that lacks context, and the removal of HTML tags, URLs, and English-only segments.
- **Ethical Considerations:** All automated data scraping will strictly comply with the robots.txt policies and terms of service of the respective news platforms, following fair-use research practices. No personal or sensitive data will be collected, minimizing ethical concerns. Furthermore, a filtering mechanism will be applied to remove profanity and culturally sensitive biases from the training corpus to prevent the model from generating harmful content.

3.3. Technical Methods and Algorithms

The development of the summarization artifact utilizes state-of-the-art Natural Language Processing (NLP) techniques tailored for low-resource languages:

- **Base Architecture:** The system uses a foundational transformer-based Large Language Model, specifically the Sinhala-adapted SinLlama (based on Llama-3-8B), which has undergone continual pre-training on millions of Sinhala tokens to deeply understand the language's morphology [1].
- **Fine-Tuning Algorithm:** Due to the severe memory bottlenecks associated with full-model training, the system employs Parameter-Efficient Fine-Tuning (PEFT), specifically utilizing Low-Rank Adaptation (LoRA) [5]. LoRA fine-tunes the model by modifying only a lightweight subset of parameters (adapters) while keeping the base pre-trained weights frozen [5]. This allows dynamic swapping of writing styles (e.g., editorial vs. breaking news) without requiring multiple heavy models.

- Tokenization: A custom Byte-Level Byte Pair Encoding (BPE) tokenizer extended with Sinhala vocabulary will be utilized to efficiently process complex Sinhala scripts and prevent fragmentation [5].
- Factual Consistency Filtering: To mitigate the model hallucination issues prevalent in abstractive summarization, the system incorporates an internal verification mechanism to ensure generated summaries do not fabricate facts from the raw input.

3.4. Tools and Platforms with Justification

The system is implemented using the following technologies:

Tool / Platform	Purpose	Justification
SinLlama	Base LLM	Sinhala-adapted decoder LLM
PyTorch	Model training	Industry-standard deep learning framework
HuggingFace Transformers	Fine-tuning & tokenizer handling	Efficient LLM experimentation
PEFT (LoRA)	Parameter-efficient adaptation	Reduces training cost (Hu, et al., 2022)
Google Colab / GPU Server	Training environment	Required for LLM fine-tuning
ROUGE, BLEU	Evaluation metrics	Standard NLP evaluation

Table 1-Tools and Platform with Justification

3.5. Validation Strategy and Performance Metrics

To prove the solution works and surpasses current standards, the model's generated summaries will be evaluated against human-generated reference summaries using both traditional and modern NLP metrics:

- ROUGE Metrics (Recall-Oriented Understudy for Gisting Evaluation): This will serve as the primary quantitative baseline [3].
 - ROUGE-1 and ROUGE-2 will measure unigram and bigram overlap, evaluating how accurately key informational phrases are captured [3].
 - ROUGE-L will measure the Longest Common Subsequence (LCS) to evaluate sentence-level structural similarity and fluency [3].
- Semantic Evaluation (LLM-as-a-Judge): Because ROUGE relies on exact word matching and struggles with abstractive paraphrasing, an advanced LLM (e.g., GPT-4o or DeepSeek) will be used to qualitatively evaluate the semantic alignment, factual consistency, and specific style retention of the summaries generated.
- Baseline Comparison: The performance will be benchmarked against existing generalized abstractive models and traditional extractive models to prove the validity of the research gap.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of Proposed Solution and its Component

The proposed solution, SinhalaJournal-LLM, is an AI-powered language assistance toolkit designed to support Sinhala newspaper writing. Rather than acting as a rigid, sequential pipeline, the system functions as a versatile suite of independent editorial tools, allowing the journalist to explicitly choose the exact operation they need. The architecture is structured into three distinct layers:

1. **Input Layer:** Handles the ingestion of raw Sinhala text and the journalist's specific formatting requirements (action and style).
2. **Process Layer (Core NLP Engine):** The central processing hub powered by a Base Large Language Model (SinLlama / Llama-3) augmented with Parameter-Efficient Fine-Tuning (PEFT). It houses three independent, parallel generation modules (Headline and Image Generator, News Summarizer, Style Rewriter) and a unified post-processing Grammar Checker.
3. **Output Layer & Feedback:** Presents the generated artifacts and facilitates a human-in-the-loop refinement cycle.

4.2. System Diagram of Components

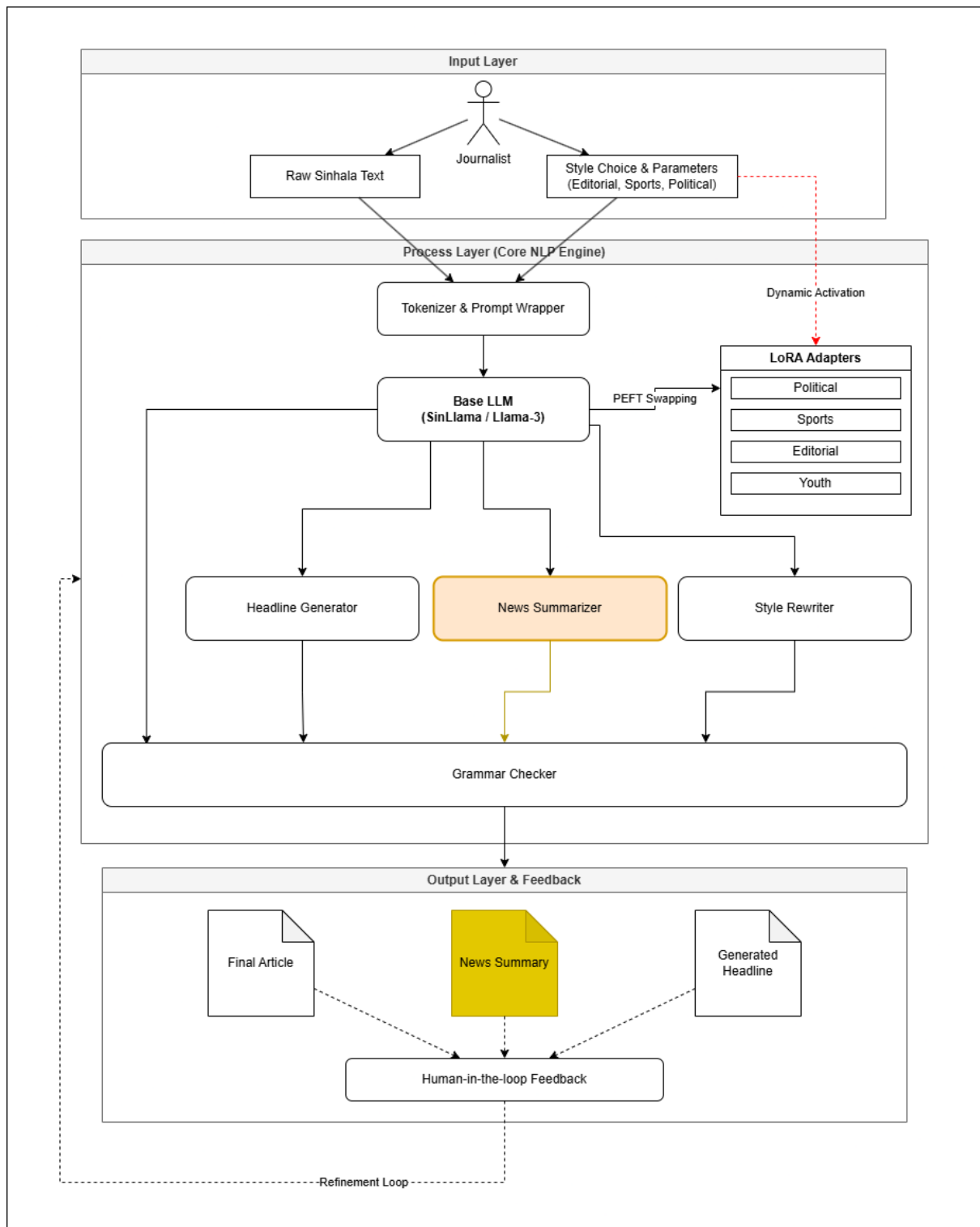


Figure 2-Sinhala Journal LLM High-Level System Architecture Diagram

4.3.Highlight the Individual Component: Sinhala News Summarizer

Operating as one of the independent, parallel tools within the Core NLP Engine, the News Summarizer focuses exclusively on condensing the tokenized text. When the journalist explicitly selects the summarization action, this component activates its internal mechanics:

- Context Window Manager: Chunks the incoming raw text to fit the LLM's maximum token limit (e.g., 8K context length) without losing critical narrative flow.
- Abstractive Generation Node: Utilizes the active state of the Base LLM combined with the dynamically loaded LoRA adapter weights (e.g., "Sports") to decode the text and generate a stylized, paraphrased Sinhala summary.
- Factual Consistency Filter: An internal validation mechanism that cross-references the newly generated abstractive summary against the original raw text to catch and mitigate AI hallucinations.

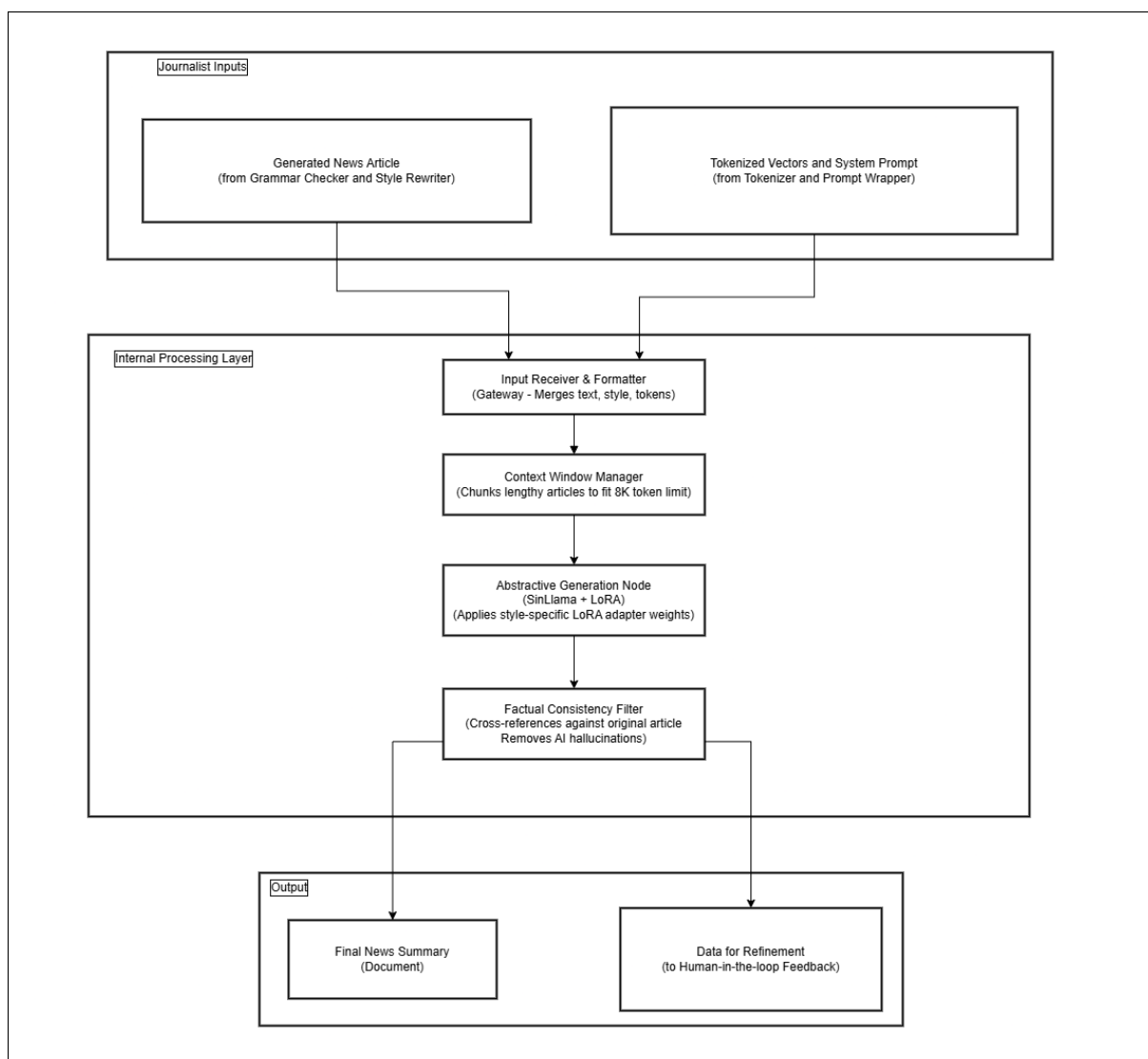


Figure 3-Sinhala Journal LLM News Summarizer Component

4.4.Component Interaction and Data Flow Explanation

The News Summarizer relies on tightly coupled interactions to provide a seamless user experience:

- **Input & Action Selection (Data Flow):** The journalist inputs a raw Sinhala news article and explicitly selects the "Summarize" action along with a specific tone (e.g., "Editorial").
- **Dynamic Adapter Loading (Interaction):** The Style Choice parameter sends a trigger to the LoRA Adapters container. The system dynamically loads the "Editorial" LoRA adapter onto the Base LLM via PEFT swapping.
- **Summarization (Interaction):** Because the user selected the summarization tool, the Base LLM ignores the Rewriter and Headline branches. Instead, it processes the tokenized vectors through the News Summarizer, generating an abstractive summary reflecting the formal tone of an editorial.
- **Grammar Validation (Data Flow):** The generated summary does not go straight to the user. Instead, it flows downward into the Grammar Checker. This unified component ensures that any morphological or syntactic errors introduced during the abstractive generation phase are corrected.
- **Output & Refinement (Data Flow):** The grammatically validated News Summary is pushed to the Output Layer. If the journalist edits the summary, this data enters the Human-in-the-loop Feedback node, flowing back via the Refinement Loop to prompt the system to adjust its output dynamically.

4.5.Scalability and Security Considerations

- **Scalability via PEFT Swapping:** Deploying multiple 8-billion parameter Large Language Models for different writing styles and tasks is computationally impossible for standard servers. The architecture achieves high scalability by keeping a single, heavy Base LLM (SinLlama) loaded in the GPU VRAM. Depending on the user's style choice, the system dynamically swaps lightweight LoRA Adapters (which are only a few megabytes) in and out of memory. This allows the system to serve multiple concurrent journalists efficiently without memory bottlenecks.
- **Data Privacy:** Journalists work with highly sensitive, unpublished scoops and drafts. The architecture guarantees that the Raw Sinhala Text is processed ephemerally in memory. It is not permanently stored in a database or used to pre-train the base model without explicit consent, protecting the media agency's intellectual property.
- **Content Integrity (Anti-Hallucination):** Abstractive LLMs inherently risk fabricating information. By implementing an internal factual consistency filter within the summarizer and routing the final output through a strict Grammar Checker, the system acts as a security guardrail against generating factually incorrect or defamatory summaries.

5. USER REQUIREMENTS

5.1. Stakeholder Identification

To ensure the system solves a genuine real-world problem, the stakeholders for the SinhalaJournal-LLM system—and specifically the News Summarizer component—have been identified based on the fast-paced nature of modern newsrooms and the demographics of the target audience:

- **Primary Stakeholders (Direct Users):** Sinhala journalists, sub-editors, and digital content creators. In today's digitalized world with its hustle and bustle, mobile users often skim and scroll, skipping lengthy articles [2]. Journalists need a tool to rapidly produce concise summaries so readers can quickly understand the main idea of the content [2].
- **Secondary Stakeholders:** Sri Lankan media institutions, online news publishers, and newspaper agencies aiming to reduce editorial bottlenecks, improve efficiency, and overcome the laborious task of manual text summarization [2].
- **Tertiary Stakeholders (Indirect Beneficiaries):** The general Sinhala-speaking public. Sinhala is the primary language for an estimated 20 million people in Sri Lanka [1], with around 11.5 million internet users [6]. Providing accurate, easily digestible summaries helps convey the most important context of a document to this wider community [6].

5.2. Functional Requirements

The following functional requirements define what the Sinhala News Summarizer must explicitly do to satisfy the primary stakeholders' needs:

- **FR1 - Input Acceptance:** The system shall allow the user to input raw, unstructured Sinhala news articles for summarization.
- **FR2 - Action and Style Selection:** The system shall provide an interface for the user to explicitly select the "Summarize" action and choose a specific journalistic domain/style (e.g., Political, Sports, Editorial). This selection will dynamically trigger the corresponding Low-Rank Adaptation (LoRA) adapter [5].
- **FR3 - Abstractive Summarization Generation:** The system shall generate a concise, abstractive summary of the inputted Sinhala text. Unlike extractive methods, the model must be able to create new words and phrases that mimic human-like summarization while retaining the original meaning [8].
- **FR4 - Grammar Validation Routing:** The generated abstractive summary shall not be outputted directly to the user; it must first be automatically routed to the system's unified Grammar Checker to ensure syntactic accuracy.
- **FR5 - Human-in-the-Loop Editing:** The system shall present the final grammatically validated summary to the user, allowing them to review, manually edit, and provide feedback on the output before final publication.

5.3. Non-Functional Requirements

The following non-functional requirements define the quality attributes, constraints, and performance metrics of the system:

- **Factual Consistency and Reliability (Anti-Hallucination):** The abstraction abilities of neural models can sometimes lead to the distortion or fabrication of factual information, causing inconsistency between the original text and the summary [8]. A critical requirement is that the system must mitigate "model hallucination" where the generated text deviates factually from the input [8]. The summarizer must maintain strict factual integrity, ensuring it does not fabricate names, numbers, or events.
- **Scalability under Hardware Constraints:** Training and deploying Large Language Models (LLMs) often face severe hardware limitations and memory bottlenecks [5]. To ensure the system is scalable within a newsroom's limited computational infrastructure, it must utilize Parameter-Efficient Fine-Tuning (PEFT) techniques like LoRA. This allows the system to achieve significant memory savings [5] by dynamically swapping lightweight adapters rather than loading multiple heavy models into the GPU simultaneously.
- **Performance (Latency):** Processing long documents with transformer-based models can be computationally expensive and less efficient due to quadratic complexity [8]. The summarizer must be optimized to process standard-length Sinhala news articles and generate the summary within an acceptable latency threshold (e.g., under 15 seconds) so as not to disrupt the fast-paced editorial workflow.
- **Security and Data Privacy:** Journalists frequently work with sensitive, unpublished drafts. The system must process input data ephemerally. Raw articles must not be permanently stored in external cloud databases or used to continuously train the base model without explicit user consent, thereby protecting the media agency's intellectual property.
- **Usability:** The user interface must be intuitive and easy to navigate for journalists who may not have a technical background in Natural Language Processing.

6. COMMERCIALIZATION PLAN

6.1. Target market and customer persona

The primary market includes Sri Lankan media institutions, online news publishers, and digital content creation agencies. The secondary market encompasses freelance journalists, journalism students, and social media content creators. There is a significant audience base, as Sinhala is the primary language for approximately 21 million people in Sri Lanka, with around 11.5 million active internet users [6].

A busy sub-editor or digital content manager in a fast-paced Sri Lankan newsroom. This professional currently relies heavily on manual processes to condense lengthy print articles into short, engaging summaries for web portals. They face strict publishing deadlines and struggle with slow editorial workflows and the high operational costs associated with manual editing.

6.2. Value proposition

Currently, manually summarizing Sinhala text is a time-consuming and laborious task [2]. Existing automated solutions for Sinhala either rely on rigid extractive algorithms (like TF-IDF or TextRank) that simply extract and copy sentences without paraphrasing [2] [7], or utilize generalized multilingual models that struggle with Sinhala grammar, factual consistency, and journalistic tone. The SinhalaJournal-LLM News Summarizer provides an AI-powered, style-aware abstractive summarization tool tailored specifically for the Sri Lankan news industry. It drastically reduces the time required for editorial workflows and lowers operational costs by instantly generating concise, human-like summaries that perfectly match the original article's specific journalistic style (e.g., sports, political, editorial).

6.3. Revenue model

The product will be commercialized as a B2B Software-as-a-Service (SaaS) platform with a tiered subscription model:

- Basic Tier (B2C/Freelancers): A low-cost monthly subscription allowing individual journalists and students to summarize a limited number of articles per month via a web interface.
- Enterprise Tier (B2B/Newsrooms): A premium, high-tier subscription for major media agencies that provides unlimited usage, multiple user accounts, and dedicated API access to seamlessly integrate the summarizer into their existing Content Management Systems (CMS).

6.4. Cost estimation

The initial development phase will utilize university resources, but commercial deployment will incur several operational costs:

The initial development phase will utilize university resources, but commercial deployment will incur several operational costs:

- Infrastructure Costs: Abstractive summarization via Large Language Models requires substantial computational power. The primary cost will be renting high-VRAM cloud GPU

instances (e.g., NVIDIA A40 with 48GB VRAM) necessary to host the base LLM and dynamically swap LoRA adapters for fast inference [5].

- **Maintenance & Data Costs:** Ongoing expenses for server upkeep, system maintenance, and continuous automated web scraping pipelines to regularly update the model's vocabulary with new Sinhala terminology [5].
- **Marketing & Onboarding:** Costs associated with pitching the software to media institutions and providing technical support or training for newsroom staff.

6.5.Pricing strategy

A Value-Based Pricing Strategy will be employed. Because the heavy dependence on manual editing significantly increases operational costs for media outlets, the enterprise subscription will be priced at a fraction of the cost of hiring an additional full-time sub-editor. This provides a highly attractive Return on Investment (ROI) for news agencies by eliminating editorial bottlenecks while saving money.

6.6.Competitive advantage

The primary competitive advantage is Domain and Language Specialization. Global multilingual Large Language Models (MultiLMs) often suffer from the "curse of multilinguality," where their limited capacity causes them to underperform on low-resource languages like Sinhala compared to high-resource languages [11]. Current generalized systems lack structural style control and cannot differentiate between the formal tone of an editorial and the condensed style of breaking news. By utilizing Parameter-Efficient Fine-Tuning (PEFT) and Low-Rank Adaptation (LoRA) on a Sinhala-optimized base model (SinLlama) [1] [5], this system possesses a unique capability: it is the only abstractive summarizer explicitly trained to understand and replicate Sri Lankan journalistic styles, offering a level of fluency and contextual accuracy that competitors cannot match.

6.7.Intellectual property considerations

- **Proprietary Assets:** The core intellectual property (IP) includes the custom-curated Instruction Fine-Tuning (IFT) datasets [5], the specialized Byte-Level BPE tokenizer configured for Sinhala [5], and the trained LoRA adapter weights for distinct writing styles [5]. These will remain closed-source for commercial enterprise deployment.
- **Data Privacy & Client IP:** Because journalists frequently work with highly sensitive, unpublished scoops, the system will guarantee that user-inputted raw texts are processed ephemerally. Client data will not be permanently stored or used to pre-train public models without explicit consent, thereby legally safeguarding the media agency's exclusive journalistic IP.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated resources required to successfully develop, train, and evaluate the SinhalaJournal-LLM News Summarizer over a 12-month project lifecycle. Because this is an academic undergraduate research project, the team leverages existing personal resources, open-source 2025 developer tools, and university infrastructure wherever possible to keep actual out-of-pocket funding requirements to a minimum.

7.1. Estimated Budget Breakdown

Category	Item Description	Estimated Market Cost	Actual Funding Required
Hardware Cost	4 Development Laptops (Local Dev)	Rs. 1,600,000.00	Rs. 0.00 (Pre-owned)
Software Cost	Open-Source NLP Stack & IDEs	Rs. 0.00	Rs. 0.00
	Commercial APIs (DeepSeek / Claude)	Rs. 30,000.00	Rs. 30,000.00
Cloud Services	Cloud GPU Instance (NVIDIA A40/A100)	Rs. 150,000.00	Rs. 150,000.00
Human Effort Cost	4 Undergraduate Researchers (12 Months)	Rs. 3,840,000.00	Rs. 0.00 (Academic Duty)
Total		Rs. 5,620,000.00	Rs. 180,000.00

Table 2-Budget Breakdown

7.2. Justification for Each Expenses

Hardware Costs:

Justification: High-performance laptops are required by the four research team members for day-to-day coding, building the custom Sinhala web scrapers, data cleaning pipelines, and drafting the research documentation. With the rising costs of computer components in 2025, a capable machine for local data processing is valued at approximately Rs. 400,000 per unit [13]. Because the team members already own these personal devices, this represents a pre-existing resource and requires zero actual funding.

Software Licenses:

Justification: The core machine learning development relies entirely on free, open-source frameworks, including Python, PyTorch, the PEFT library, and Hugging Face Transformers [5]. Furthermore, development will be conducted using free, industry-standard IDEs such as Visual Studio Code, which offers robust Python and AI integration out of the box at no cost.

API Credits (Funding Requested): A small budget of Rs. 30,000 is allocated for commercial LLM API access. As outlined in the methodology, a modern "LLM-as-a-Judge" approach will be utilized alongside traditional metrics to evaluate the semantic alignment and style retention of the generated Sinhala summaries using advanced models like DeepSeek or Claude [5].

Cloud Services (Crucial Expense):

Justification: This is the primary out-of-pocket expense for the project. The Sinhala News Summarizer utilizes a foundational 8-billion parameter model (SinLlama / Llama-3). Fine-tuning an LLM of this scale—even when utilizing Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA—requires significant GPU Video RAM (VRAM) that standard laptops cannot support [5]. Current research indicates that high-end GPUs such as the NVIDIA A40 (with 48GB VRAM) or A100 are necessary to prevent memory bottlenecks during continued pre-training and tuning [5]. Rather than purchasing physical GPUs—which face severe pricing premiums in 2025 [13] cloud compute platforms (e.g., RunPod, Lambda Labs, or AWS) will be rented on an hourly basis. A budget of Rs. 150,000 is allocated to cover the intensive training, hyperparameter tuning, and validation phases across the 12 months [5].

Human Effort Cost:

Justification: Developing a style-aware abstractive summarizer for a low-resource language like Sinhala requires intensive, specialized manual labor. This includes writing custom scraping scripts, rigorously cleaning noisy Sinhala text, and conducting human-in-the-loop qualitative evaluations over the 12-month period. According to the Software Engineer Salary Guide: Sri Lanka 2025, the current market rate for entry-level software engineers ranges from Rs. 80,000 to Rs. 150,000 per month [14]. Valuing the four team members' highly technical NLP effort at the baseline of Rs. 80,000 per month, the total estimated labor cost is Rs. 3,840,000. However, as this represents the students' academic time investment toward their degree, it requires no actual financial payout.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1. Overview of Work Breakdown

The project is divided into the following major phases:

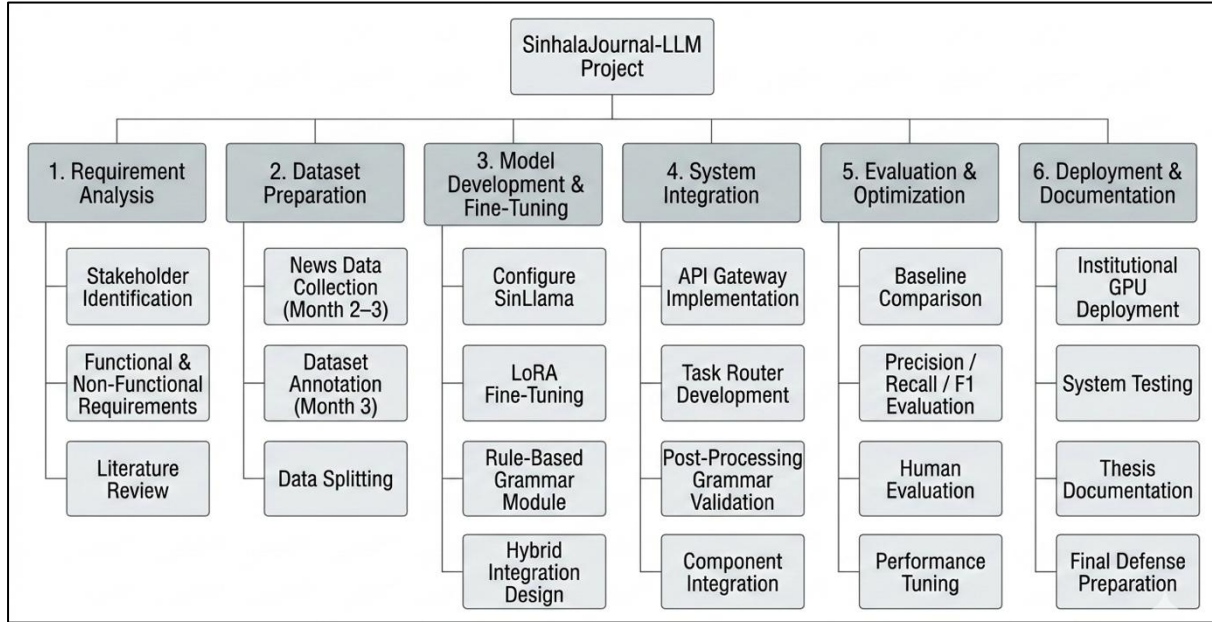


Figure 4-Work Breakdown Structure (WBS)

Figure 4 presents the hierarchical Work Breakdown Structure (WBS) of the SinhalaJournal-LLM project. The project is organized into six major phases with clearly defined tasks to ensure structured development across the one-year timeline. The Sinhala Grammar Checker activities—LoRA fine-tuning, rule-based validation, and hybrid integration—are highlighted to reflect the individual research contribution. The WBS also demonstrates balanced workload distribution and coordinated system-level integration.

8.2. Timeline Alignment

The project spans twelve months, following a phased and progressive development schedule:

- Months 1–2: Requirement analysis and literature review
- Months 2–3: Dataset collection
- Month 3: Dataset annotation
- Months 4–6: SinLlama fine-tuning and rule-based module development
- Months 6–7: Hybrid integration and system integration
- Months 7–9: Experimental evaluation and optimization
- Months 9–10: Institutional deployment and system testing

- Months 7–12: Documentation and thesis writing
- Months 10–12: Final validation and defense preparation

This timeline allows iterative experimentation, refinement, and integration while maintaining feasibility within the academic year.

8.3.Task Breakdown per Member

Phase	Member 1 – Grammar Checker	Member 2 – Headline Generator	Member 3 – Style Rewriter	Member 4 – News Summarizer
Requirement Analysis	Define objectives and evaluation metrics	Define generation objectives and style criteria	Define transformation objectives and tone categories	Define summarization objectives and evaluation criteria
Dataset Preparation	Prepare annotated error-correction dataset	Prepare labeled headline dataset	Prepare style transformation dataset	Prepare summarization dataset
Model Development	Implement LoRA fine-tuning and rule-based validation	Implement style-conditioned fine-tuning	Implement instruction-based fine-tuning	Implement summarization model
System Integration	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router

8.4.Balanced Workload Distribution

Workload distribution is structured to ensure fairness and technical parity among team members. While the Sinhala News Summarizer involves designing context window management, training an abstractive LoRA adapter, and implementing a factual consistency filter, other components require comparable effort in dataset preparation, style adaptation, and rule-based development. Shared tasks and specific technical lead roles—such as managing the data pre-processing pipelines, configuring the custom Byte-Level BPE tokenizer, and overall system testing—further balance the technical contribution. This structured allocation ensures that each member contributes meaningfully to both individual research objectives and overall platform development.

9. GANTT CHART

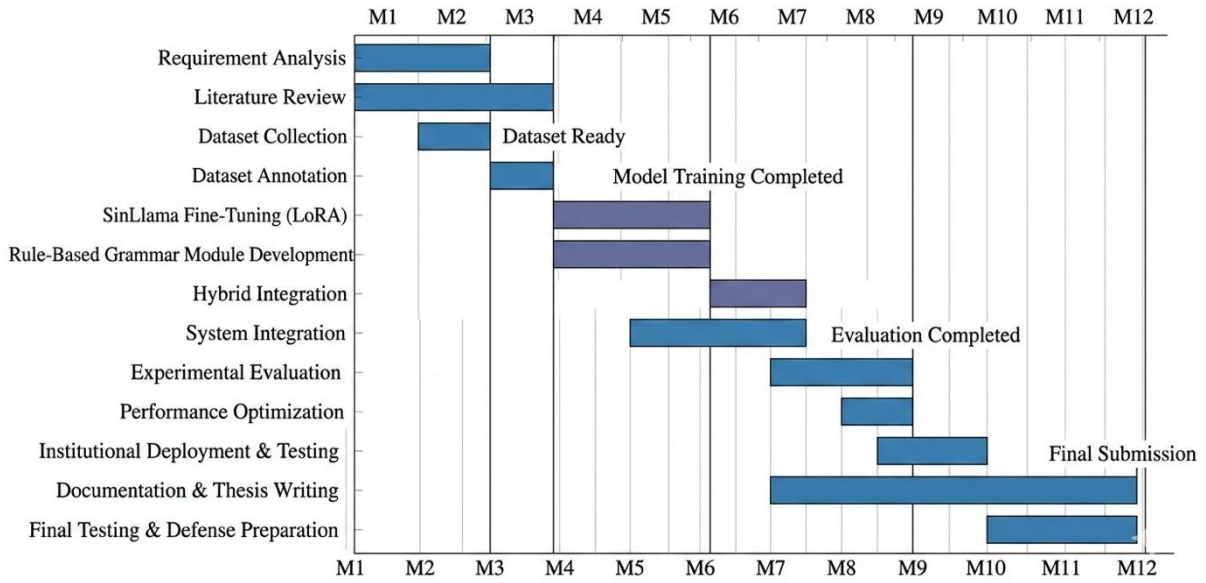


Figure 5-12 Month project timeline grantt chart

Figure 5 presents the 12-month project timeline in the form of a Gantt chart. The schedule aligns research, development, integration, and evaluation phases in a structured and progressive manner. Early months focus on requirement analysis and dataset preparation, followed by model fine-tuning and rule-based module development. System integration and experimental evaluation occur in the middle phase, while deployment, documentation, and final validation are scheduled in the concluding months.

Milestones are clearly defined to monitor progress, including dataset completion, model training completion, evaluation finalization, and final submission. The timeline ensures feasibility, iterative refinement, and sufficient preparation for project defense.

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11. APPENDICES

11.1. Preliminary UI mockups

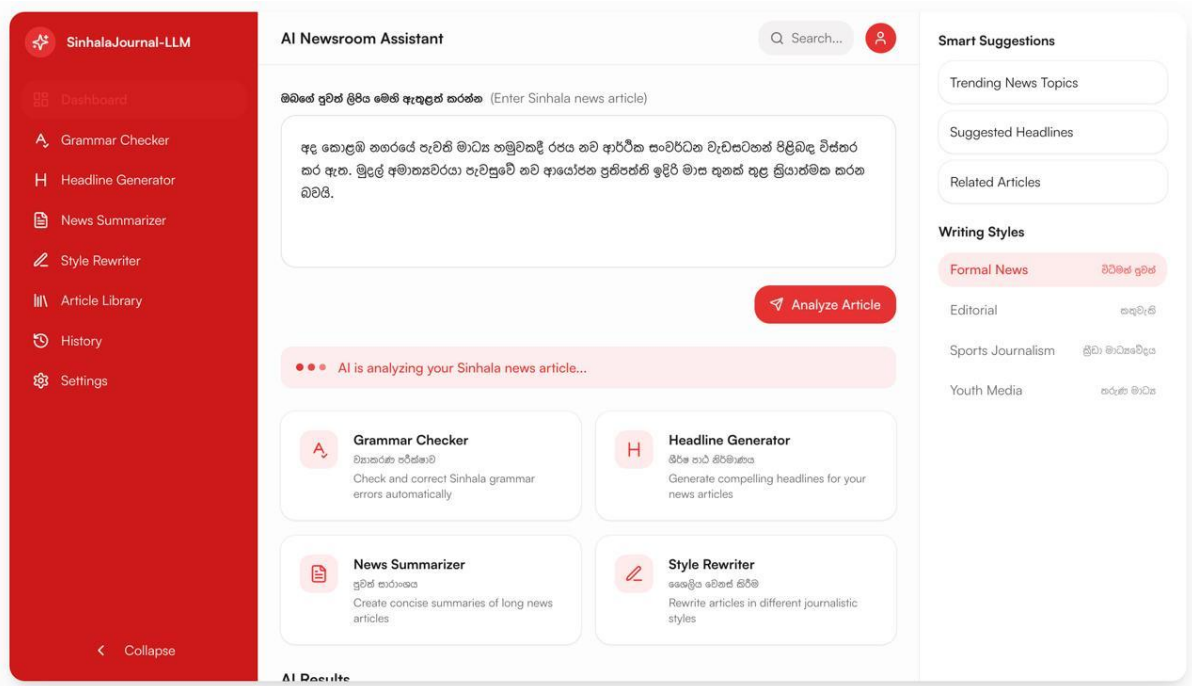


Figure 6 Proposed Web Interface for SinhalaJournal-LLM showcasing the split-screen generation and human-in-the-loop refinement loop

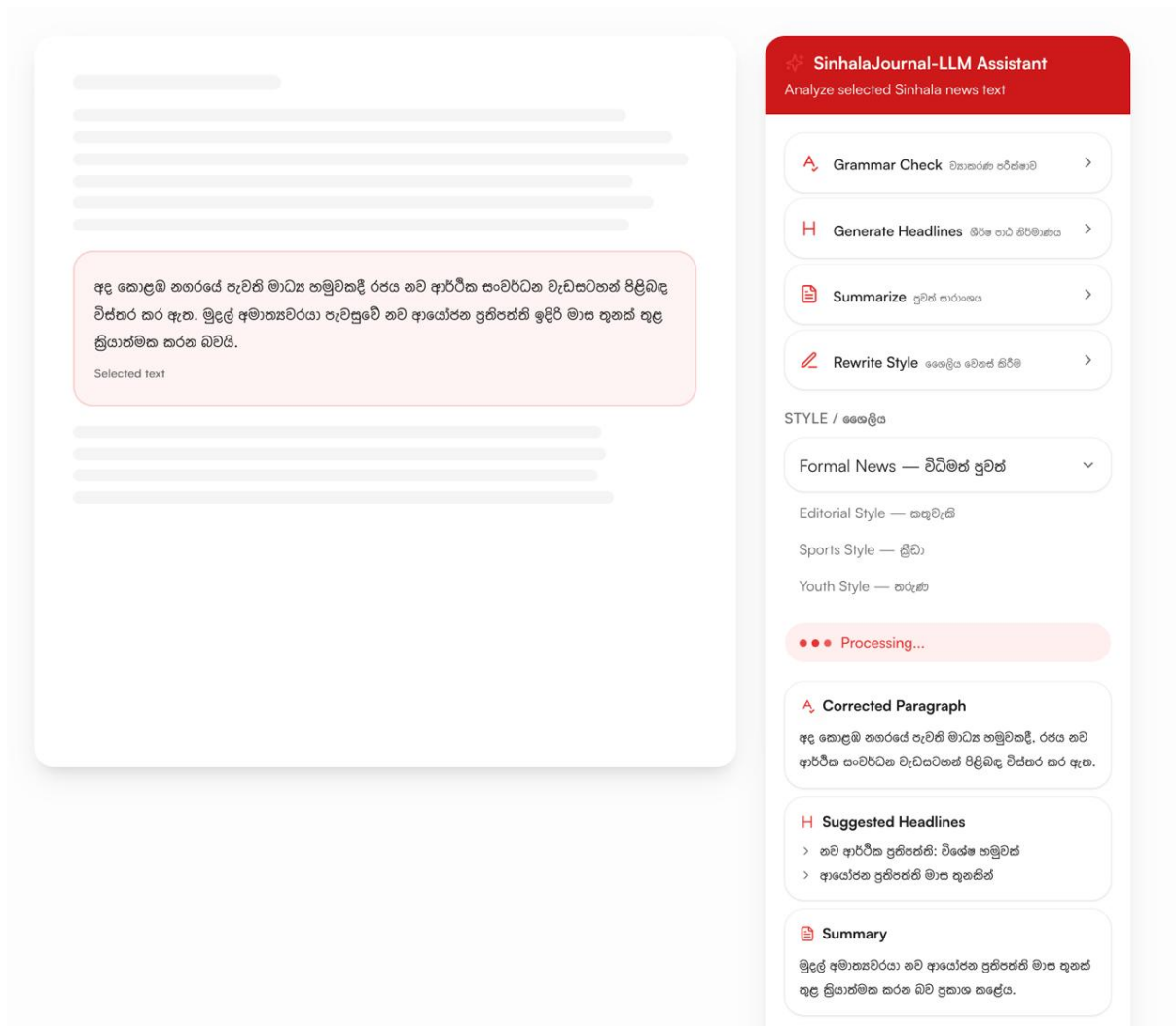


Figure 7-Proposed Google Doc Plugin Interface

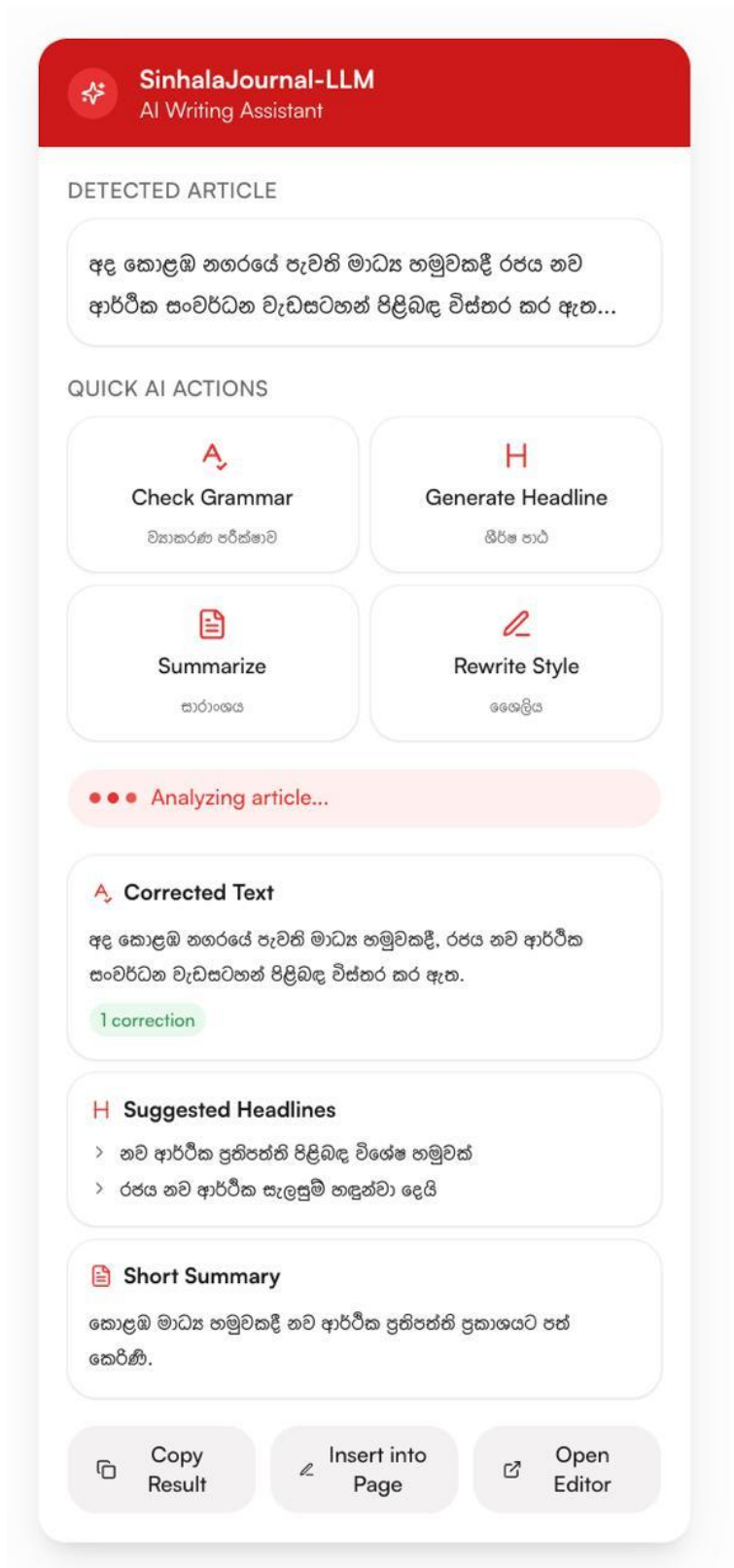


Figure 8-Proposed Browser Extension allowing journalists to seamlessly summarize and rewrite text directly from online sources

11.2. Risk analysis

Risk ID	Risk Category	Risk Description	Probability	Impact	Mitigation Strategy / Contingency Plan
R1	System Output	Model Hallucination & Factual Inconsistency: Abstractive summarization models inherently risk fabricating factual information (names, numbers, events) or distorting the original meaning of the text. This is a critical risk in journalism.	High	Critical	Design and integrate an internal Factual Consistency Filter . This validation mechanism will actively cross-reference the generated abstractive summary against the original raw text to catch and eliminate hallucinations before the output is passed to the Grammar Checker.
R2	Technical / Hardware	High Computational Complexity: Transformer-based models (like SinLlama/Llama-3-8B) suffer from quadratic complexity with respect to input text length. Processing lengthy Sinhala news articles risks severe GPU memory bottlenecks and system crashes.	High	Critical	Implement Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA) to drastically reduce trainable parameters and memory footprint. Additionally, integrate a Context Window Manager to chunk long articles to fit the 8K token limit without losing narrative flow. Rent high-VRAM Cloud GPUs (e.g., NVIDIA A40) for training.
R3	Data	Low-Resource Language Data Scarcity: Sinhala is an extremely low-resource language. There is a severe lack of readily available, domain-annotated, large-scale Sinhala abstractive summarization datasets	High	High	Curate a custom dataset by deploying automated web scrapers to gather raw articles from established Sri Lankan news platforms (e.g., Lankadeepa, Ada, Hiru News, Divaina). Apply rigorous data pre-

		required for style-aware training.			processing to clean noise and remove HTML tags.
R4	Evaluation	Limitations of ROUGE Metrics: Traditional ROUGE metrics primarily measure exact n-gram overlap. Because abstractive models generate new words and paraphrase text, ROUGE may inaccurately penalize valid, semantically correct, human-like summaries.	Medium	Medium	Adopt a hybrid validation strategy. While ROUGE-1, 2, and L will be used as quantitative baselines, the system will primarily rely on modern " LLM-as-a-Judge " (e.g., DeepSeek, Claude) qualitative evaluations to accurately assess semantic meaning, factual consistency, and specific style retention.
R5	Generation Control	Summary Length Control: Abstractive models often struggle to produce summaries of an exact, desired length (e.g., fitting a specific newspaper column slot) while maintaining coherence.	Medium	Medium	Engineer optimized prompt-wrappers that provide explicit length and style constraints to the base LLM during generation. The Context Window Manager will also assist in regulating the size of the input chunks processed at any given time.
R6	Ethical & Legal	Data Privacy Breaches: The summarizer may process highly sensitive, unpublished journalistic scoops, risking intellectual property theft if the data is retained.	Low	High	Enforce a strict ephemeral memory processing policy. Raw articles inputted by journalists will be processed in-memory and immediately discarded. They will not be permanently stored in cloud databases or used for continual model pre-training.

Table 3-Risk Assessment and Mitigation Matrix for the News Summarizer

11.3. Ethical documents

Page 2 of 38 - Integrity Overview

Submission IDtrn:oid::3618:131348677

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Match Groups

35 Not Cited or Quoted7%

Matches with neither in-text citation nor quotation marks

12 Missing Quotations2%

Matches that are still very similar to source material

0 Missing Citation0%

Matches that have quotation marks, but no in-text citation

0 Cited and Quoted0%

Matches with in-text citation present, but no quotation marks

Top Sources

5%

Internet sources

3%

Publications

7%

Submitted works (Student Papers)